Proper Time and Kinetic Energy in Special Relativity: A Rate Expansion and an Exact Identity

Light-Meter Physics

Abstract

For a clock moving through flat spacetime, proper time and the coordinate time of a chosen inertial frame are related by $\mathrm{d}\tau/\mathrm{d}t = \sqrt{1 - v^2/c^2} = 1/\gamma$. Expanding this rate about zero speed gives a convergent series whose corrections are all negative. A separate but exact energy identity shows that the fractional reduction of the clock rate equals the kinetic-energy fraction of the particle's total energy in that same frame:

$$1 - \frac{\mathrm{d}\tau}{\mathrm{d}t} = \frac{K}{E} = 1 - \frac{1}{\gamma}.$$

The two fractions share the same leading Newtonian term but differ at higher orders when kinetic energy is instead normalized by rest energy. A worked example, the proper-time integral for variable speed, and a discussion of notation delimit the result and its interpretation. The analysis is kinematic: it does not invoke gravitational time dilation.

Keywords: proper time, time dilation, Lorentz factor, relativistic kinetic energy, binomial expansion

1 Introduction

Proper time is the time recorded by an ideal clock along its own worldline. In special relativity, its relation to a coordinate time depends on the chosen inertial frame and the clock's speed in that frame. This relation is elementary but connects naturally to an exact energy statement: for a massive particle, the fraction of its total energy assigned to kinetic energy equals the fractional reduction of its proper-time rate relative to that frame's coordinate time.

This manuscript derives the relation from the flat-spacetime interval, expands the rate in powers of v/c, and distinguishes the ratio K/E from $K/(mc^2)$, which has a different

higher-order expansion. The standard binomial series supplies the coefficients [1], while MIT OpenCourseWare's notes on spacetime kinematics and four-momentum provide a broader special-relativistic context [2].

The word "reduction" below refers only to a comparison of rates, $1-\mathrm{d}\tau/\mathrm{d}t$, with respect to one selected inertial frame. It does not mean that the moving clock locally runs abnormally: the clock measures its own proper time in its instantaneous rest frame. Nor does the identity claim that kinetic energy causes time dilation; both expressions depend on the same Lorentz factor.

2 Proper-time rate in an inertial frame

In flat spacetime, the interval along a timelike worldline, using inertial coordinates $(t, \mathbf{x})$ and metric signature $(-,+,+,+)$, is

$$\begin{aligned} \mathrm{d}s^2 &= -c^2\,\mathrm{d}\tau^2 \\ &= -c^2\,\mathrm{d}t^2 + \mathrm{d}\mathbf{x}\cdot\mathrm{d}\mathbf{x}. \end{aligned} \tag{1}$$

Here $\mathrm{d}\tau$ is the positive proper-time increment of a future-directed clock. Let

$$\mathbf{v} \equiv \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t}, \qquad v \equiv |\mathbf{v}|, \qquad \beta \equiv \frac{v}{c}, \qquad \gamma \equiv \frac{1}{\sqrt{1-\beta^2}}. \tag{2}$$

Dividing Eq. (1) by $c^2\mathrm{d}t^2$ and taking the positive root gives

$$\boxed{\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{1-\frac{v^2}{c^2}} = \frac{1}{\gamma}.} \tag{3}$$

The relation is exact for any timelike motion in flat spacetime, including accelerated motion, provided v is evaluated instantaneously in the same inertial coordinate frame. For a massive clock, $0 \le v < c$.

Equation (3) compares two different elapsed-time measures: the clock's proper time and the coordinate time in the selected frame. It is not a statement that the local laws of physics or the clock mechanism change in the clock's own rest frame. Although τ is invariant along a specified worldline, the coordinate time t, the speed v, and the ratio $\mathrm{d}\tau/\mathrm{d}t$ depend on the chosen inertial frame.

3 Series expansion of the rate

Introduce the dimensionless expansion variable

$$x \equiv \frac{v^2}{c^2} = \beta^2, \qquad 0 \le x < 1. \tag{4}$$

The rate function is $f(x) = (1-x)^{1/2}$. Applying the generalized binomial expansion [1] gives

$$(1-x)^{1/2} = 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} - \frac{5x^4}{128} - \frac{7x^5}{256} - \cdots, \qquad |x| < 1. \tag{5}$$

Restoring the speed variable,

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = 1 - \frac{1}{2}\left(\frac{v}{c}\right)^2 - \frac{1}{8}\left(\frac{v}{c}\right)^4 - \frac{1}{16}\left(\frac{v}{c}\right)^6 - \frac{5}{128}\left(\frac{v}{c}\right)^8 - \cdots. \tag{6}$$

Every nonconstant coefficient is negative. Equivalently, define the fractional rate reduction

$$R \equiv 1 - \frac{\mathrm{d}\tau}{\mathrm{d}t}. \tag{7}$$

Then

$$R = \frac{x}{2} + \frac{x^2}{8} + \frac{x^3}{16} + \frac{5x^4}{128} + \frac{7x^5}{256} + \cdots. \tag{8}$$

The correction terms in the rate are all negative; the corresponding terms in the rate reduction are all positive.

For integer $n \ge 1$, the magnitude of the coefficient of x^n in the correction to the rate is

$$a_n \equiv \frac{(2n-3)!!}{2^n n!}, \qquad (-1)!! \equiv 1, \tag{9}$$

so that

$$\sqrt{1-x} = 1 - \sum_{n=1}^{\infty} a_n x^n. \tag{10}$$

The power series has radius of convergence one. All physical speeds of a massive clock satisfy $x < 1$, so the full infinite series converges throughout the physical domain. It also converges at the mathematical endpoint $x = 1$ and sums there to zero, but $x = 1$ means $\beta = 1$: a null worldline, not a possible trajectory for a massive clock. For finite truncations, the omitted terms are positive in the series for R; correspondingly, truncating Eq. (6) overestimates the exact rate.

4 An exact identity with kinetic energy

For a particle of fixed rest mass $m > 0$ in the same inertial frame, the total energy and kinetic energy are

$$E = \gamma mc^2, \qquad K \equiv E - mc^2 = (\gamma - 1)mc^2. \tag{11}$$

The rest-energy fraction of the total energy is

$$\frac{mc^2}{E} = \frac{1}{\gamma}. \tag{12}$$

Subtracting this fraction from unity gives the kinetic-energy fraction:

$$\frac{K}{E} = \frac{E - mc^2}{E} = 1 - \frac{mc^2}{E} = 1 - \frac{1}{\gamma}. \tag{13}$$

Combining Eqs. (3) and (13) yields the exact identity

$$\boxed{\frac{K}{E} = 1 - \frac{\mathrm{d}\tau}{\mathrm{d}t}.} \tag{14}$$

Thus the dimensionless kinetic-energy fraction equals the fractional reduction in proper-time rate relative to the selected inertial frame. Equivalently, the rest-energy fraction mc^2/E equals the proper-time rate $\mathrm{d}\tau/\mathrm{d}t$. Both statements follow directly from $E = \gamma mc^2$ and do not introduce a new dynamical law.

4.1 Why the denominator matters

Normalizing kinetic energy by rest energy gives a different quantity:

$$\frac{K}{mc^2} = \gamma - 1. \tag{15}$$

Its series is

$$\frac{K}{mc^2} = \frac{x}{2} + \frac{3x^2}{8} + \frac{5x^3}{16} + \frac{35x^4}{128} + \cdots, \tag{16}$$

whereas the exact kinetic fraction in Eq. (14) is

$$\frac{K}{E} = \frac{x}{2} + \frac{x^2}{8} + \frac{x^3}{16} + \frac{5x^4}{128} + \cdots. \tag{17}$$

Both quantities begin with $x/2$. Only K/E equals the rate reduction exactly; the two normalizations differ beyond leading order.

5 A numerical example

Consider constant speed $v = 0.6c$. Then

$$\beta = 0.6, \qquad \gamma = \frac{1}{\sqrt{1 - 0.6^2}} = 1.25, \qquad \frac{\mathrm{d}\tau}{\mathrm{d}t} = 0.80. \tag{18}$$

The energy fractions are

$$\frac{mc^2}{E} = 0.80, \qquad \frac{K}{E} = \frac{0.25mc^2}{1.25mc^2} = 0.20. \tag{19}$$

The proper-time-rate reduction is $1 - 0.80 = 0.20$, exactly the kinetic-energy fraction. Over a coordinate-time interval of 10 s in this frame, the clock accumulates

$$\Delta\tau = \frac{\Delta t}{\gamma} = \frac{10\,\mathrm{s}}{1.25} = 8\,\mathrm{s}. \tag{20}$$

The proper time accumulated along the specified worldline is invariant. The coordinate interval is assigned by the selected inertial frame; their numerical difference is a comparison in that frame, not a separate time interval accumulated by the clock.

6 Changing speed and accumulated proper time

If the clock's speed varies with coordinate time in a single inertial frame, integrate the instantaneous rate:

$$\Delta\tau = \tau(t_2) - \tau(t_1) = \int_{t_1}^{t_2} \sqrt{1 - \frac{v(t)^2}{c^2}}\,\mathrm{d}t. \tag{21}$$

The rate-energy identity also holds instantaneously for the particle, so along the same history

$$\Delta\tau = \int_{t_1}^{t_2} \left(1 - \frac{K(t)}{E(t)}\right)\,\mathrm{d}t, \tag{22}$$

where $E(t)$ and $K(t)$ are evaluated from the instantaneous speed in that frame. If an external agent changes the particle's speed, its energy need not be constant. Equation (22) is therefore an integral of the instantaneous identity, not a claim that total proper time is determined by one endpoint energy ratio.

7 Notation and scope

For comparison, define the dimensionless reciprocal-rate notation

$$\delta\tau_{\mathrm{kin}} \equiv \frac{\mathrm{d}t}{\mathrm{d}\tau} = \gamma, \qquad \delta\tau_{\mathrm{kin}}^{-1} = \frac{\mathrm{d}\tau}{\mathrm{d}t}. \tag{23}$$

The notation is nonstandard: $\delta\tau_{\mathrm{kin}}$ is a dimensionless reciprocal rate, not a proper-time interval, an elapsed-time difference, or an additive correction to τ. The standard Lorentz-factor notation γ is used in the derivation. With this convention, Eq. (14) can also be written $K/E = 1 - \delta\tau_{\mathrm{kin}}^{-1}$.

The result concerns kinematic time dilation in flat spacetime. It assumes energy and speed are evaluated in the same inertial frame and does not include gravitational redshift or gravitational clock-rate effects. The coordinate-time rate is frame dependent; proper time along a specified worldline is invariant. In a gravitational field, reference energy and a moving clock's inverse rate depend differently on the lapse and local motion, so the flat-spacetime identity does not carry over unchanged. The identity is an exact algebraic relation between two dimensionless quantities, not a claim that energy loss generates elapsed time.

8 Conclusion

Starting from the Minkowski interval gives the exact clock-rate relation $\mathrm{d}\tau/\mathrm{d}t = 1/\gamma$. Its expansion in $x = (v/c)^2$ has negative corrections to the rate and positive terms in the rate reduction. For a massive particle, the exact total energy $E = \gamma mc^2$ implies that the kinetic-energy fraction K/E equals this rate reduction at every subluminal speed in the same inertial frame.

The leading terms of K/E and $K/(mc^2)$ agree in the Newtonian limit, but the higher-order terms differ because the denominators are different. For variable speed, the proper time is obtained by integrating the instantaneous rate along the worldline. These results are special-relativistic kinematics; they do not extend by themselves to gravitational time dilation.

References

[1] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds., *NIST Digital Library of Mathematical Functions*, Section 4.6(ii), Eq. 4.6.7, National Institute of Standards and Technology, https://dlmf.nist.gov/4.6#E7.

[2] S. A. Hughes, *Introduction to Relativity and Spacetime Physics*, Fall 2024, MIT 8.033, Lectures 6–8: "Kinematics in Spacetime," "4-Momentum and 4-Velocity," and "Using 4-Momentum," Massachusetts Institute of Technology OpenCourseWare, lecture notes.